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A Motivation Towards Networks

With some Illustrative Examples

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Outline

1. Methodology: From Examples to Theory
2. Markov Chains and Stationary Distributions
3. Introduction to Networks
4. PageRank and Eigenvector Centrality
5. Epidemic Spreading on Networks

A Guiding Philosophy

The Central Question

How do quantities spread, stabilise, or vanish on interconnected systems?

We follow a deliberate arc:

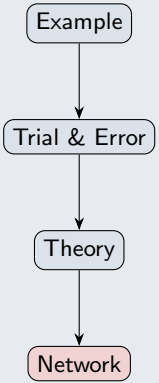
1. **Start with a concrete example.** Real phenomena ground our intuition.
2. **Trial and error.** Solve small, special cases by hand—spot the pattern.
3. **Choose the right theory.** Here: Markov chains and their algebra.
4. **Identify the network.** Strip away context; only the graph structure remains.
5. **Apply broadly.** The same theory reappears in epidemics, opinion dynamics, PageRank, ...

Roadmap

Part I (this talk)

- Population flow → Markov chains
- Stationary distributions (theory)
- What is a network?
- Epidemic spreading on networks

Recurring theme

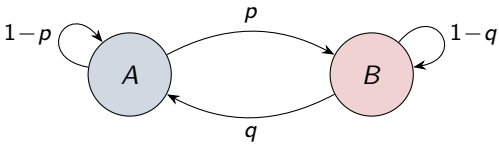


The Motivating Example: Population Flow

Setup. Two regions, A and B, with a fixed total population. Each year:

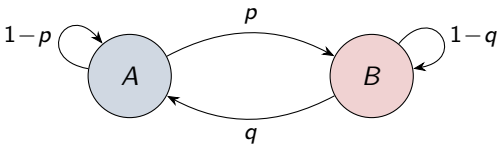
- A fraction p of A's population moves to B.
- A fraction q of B's population moves to A.
- The remainder ($1-p$ and $1-q$) stay put.

The system runs year after year, indefinitely.



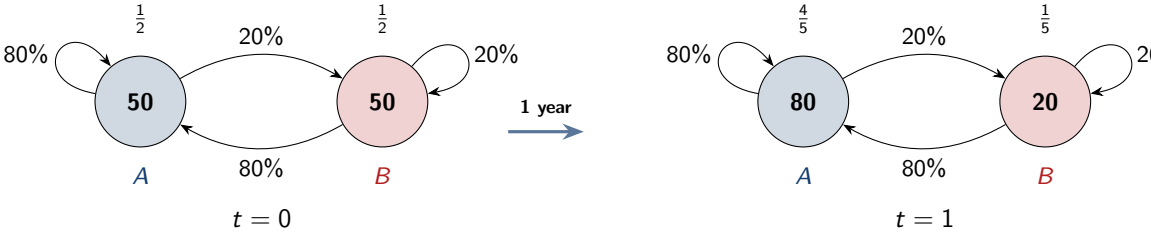
What Do You Expect?

Given this simple rule, what do you think happens to the population fractions *over many years*?



- (A) The fractions keep fluctuating — no pattern.
- (B) One region eventually absorbs *everyone*.
- (C) The system is chaotic — highly sensitive to the initial split.
- (D) **The fractions settle to a stable equilibrium.**
- (E) It is impossible to tell without more information.

Watching the System Evolve: $p = 20\%$, $q = 80\%$



$$A_1 = 50\cdot(1-0.2) + 50\cdot 0.8 = 40 + 40 = 80, \quad B_1 = 50\cdot 0.2 + 50\cdot(1-0.8) = 10 + 10 = 20.$$

The system reaches equilibrium in a single step — and stays there.

Does the Equilibrium Depend on the Initial Split?

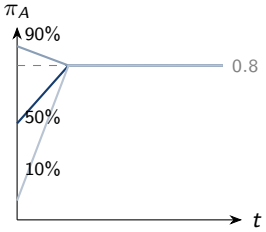
We started 50/50. But what if we had started 90/10? Or 10/90?

Try it: starting from $A_0 = 90$, $B_0 = 10$:

$$A_1 = 90{\cdot}0.8 + 10{\cdot}0.8 = 80, \quad B_1 = 20.$$

Starting from $A_0 = 10$, $B_0 = 90$:

$$A_1 = 10{\cdot}0.8 + 90{\cdot}0.8 = 80, \quad B_1 = 20.$$



Three different starting points, same equilibrium.

Observation

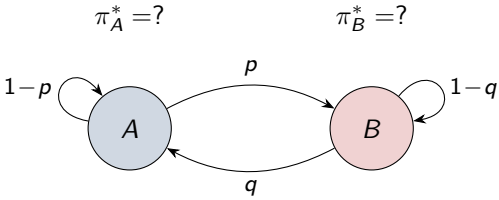
The equilibrium 80%/20% is reached *regardless* of where we start. The long-run fractions are a property of p and q alone.

Can We Calculate the Equilibrium?

We have a feeling *that* an equilibrium exists and is unique. Lets put aside the proof and ask first if we can calculate it.

Question: given p and q, can we find the equilibrium fractions (π_A^*, π_B^*) *directly* — without simulating year by year?

What conditions must π_A^* and π_B^* satisfy?



Hint

If the fractions are at equilibrium, applying the dynamics for one more year must leave them *unchanged*.

Step 1 — The Simple Case ($n = 2$): Deriving the Equilibrium

What does "stationary" mean? A distribution (π_A, π_B) is *stationary* if one year of dynamics leaves it unchanged. Let us write out what one year does to π_A:

The fraction in A *after* one step comes from two sources:

- **Those who were in A and stayed:** fraction $1-p$ of π_A.
- **Those who were in B and moved to A:** fraction q of π_B.

So the new fraction in A is

$$\pi'_A = (1-p)\,\pi_A + q\,\pi_B.$$

Stationarity demands $\pi'_A = \pi_A$, i.e.

$$\pi_A - (1-p)\pi_A = q\,\pi_B \implies p\,\pi_A = q\,\pi_B.$$

Flow balance

The condition $p\,\pi_A = q\,\pi_B$ says:

$$\textit{flow out of } A = \textit{flow into } A.$$

The population leaving A each year ($p\pi_A$) exactly replaces the population arriving from B ($q\pi_B$). No net drift occurs.

Together with $\pi_A + \pi_B = 1$:

$$\boldsymbol{\pi}^* = \left(\frac{q}{p+q}, \frac{p}{p+q}\right).$$

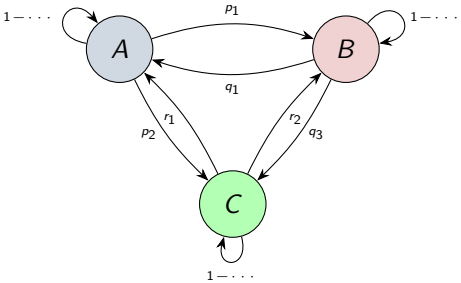
Step 2 — Three Regions: Enter the Matrix

With three regions the balance equations balloon. But notice: the transition rule can be written compactly as a **matrix** P, where

$$P_{ij} = \text{fraction moving from region } i \text{ to } j.$$

$$P = \begin{pmatrix} 1-p_1-p_2 & p_1 & p_2 \\ q_1 & 1-q_1-q_3 & q_3 \\ r_1 & r_2 & 1-r_1-r_2 \end{pmatrix}$$

Each row sums to 1: all residents end up *somewhere*.



If $\boldsymbol{\pi}^{(t)}$ is the row vector of fractions at time t, then

$$\boldsymbol{\pi}^{(t+1)} = \boldsymbol{\pi}^{(t)} P.$$

Stationarity is simply: $\quad \boldsymbol{\pi}^* P = \boldsymbol{\pi}^*, \quad \boldsymbol{\pi}^* \mathbf{1} = 1.$

Step 2 (cont.) — What If Some Connections Are Missing?

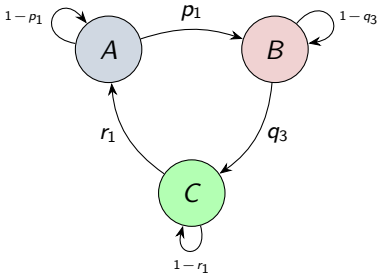
Suppose only three directed links exist:

$$A \xrightarrow{p_1} B \xrightarrow{q_3} C \xrightarrow{r_1} A.$$

(Each node also retains a self-loop for those who stay.)

The matrix becomes **sparse** — most off-diagonal entries are zero:

$$P = \begin{pmatrix} 1-p_1 & p_1 & 0 \\ 0 & 1-q_3 & q_3 \\ r_1 & 0 & 1-r_1 \end{pmatrix}$$



Note

Strip away the self-loops and the weights: the pattern of zeros and nonzeros in P is exactly the **adjacency matrix** of the directed graph on $\{A, B, C\}$. The transition matrix is a *row-normalised* adjacency matrix.

Step 3 — The General Structure

Markov Chain

A *(discrete-time, finite) Markov chain* is a random process on states $\{1, \ldots, n\}$ with *transition matrix* $P \in \mathbb{R}^{n \times n}$:

$$P_{ij} = \Pr(\text{move to } j \mid \text{currently at } i) \geq 0, \qquad \sum_j P_{ij} = 1 \; \forall \, i.$$

Key Properties

- **Irreducible**: every state reachable from every other (strongly connected graph).
- **Aperiodic**: no "locking" into cycles.
- Together: *ergodic*.

Perron–Frobenius Theorem

For a non-negative irreducible matrix P, the largest eigenvalue $\rho(P)$ is positive, simple, and has a unique (up to scale) strictly positive eigenvector.

The Main Result

Theorem (Ergodic Markov Chains)

Let P be the transition matrix of an **irreducible and aperiodic** Markov chain on n states. Then:

1. There exists a **unique** stationary (probability) distribution $\boldsymbol{\pi}^* \in \mathbb{R}^n_{>0}$ satisfying
$$\boldsymbol{\pi}^* P = \boldsymbol{\pi}^*, \qquad \sum_i \pi_i^* = 1.$$
2. $\boldsymbol{\pi}^*$ is the **left eigenvector** of P for eigenvalue $1 = \rho(P)$, normalised to unit ℓ^1 norm.
3. For *any* initial distribution $\boldsymbol{\pi}^{(0)}$,
$$\boldsymbol{\pi}^{(0)} P^t \xrightarrow{t\to\infty} \boldsymbol{\pi}^*.$$

Proof sketch. Since P is row-stochastic, 1 is always an eigenvalue. Irreducibility $\Rightarrow$ Perron–Frobenius gives uniqueness and positivity. Aperiodicity ensures all other eigenvalues satisfy $|\lambda| < 1$, so P^t converges geometrically.

What Markov Chains Reveal

- The *asymptotic* behaviour of a stochastic flow is encoded entirely in the **spectral structure** of P.
- The steady state $\boldsymbol{\pi}^*$ **balances flows**:

$$\pi_i^* P_{ij} = \pi_j^* P_{ji} \quad \text{(detailed balance, if it holds)}$$

- Removing the probabilistic framing: P is just a **weighted directed graph** — a *network*.

Markov chains $\longleftrightarrow$ weighted directed networks.

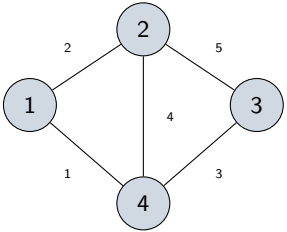
Graphs and Networks — Basic Vocabulary

A network (graph) $G = (V, E)$

- $V = \{1, \ldots, n\}$: **vertices** (nodes, agents, sites)
- $E \subseteq V \times V$: **edges** (links, connections)
- **Undirected**: $(i, j) \in E \Leftrightarrow (j, i) \in E$
- **Directed**: (i, j) and (j, i) may differ

Weights

A map $w : E \to \mathbb{R}_{>0}$ gives a **weighted network** $G = (V, E, w)$.
Weights encode strength, capacity, frequency, ...



Weighted undirected graph on 4 vertices.

The Adjacency Matrix

The **adjacency matrix** $A \in \mathbb{R}^{n \times n}$:

$$A_{ij} = \begin{cases} w_{ij} & \text{if } (i,j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

- Symmetric for undirected graphs.
- Encodes the full network structure.
- Eigenvalues of A carry rich information.

Connection to Markov chains:

$$P_{ij} = \frac{A_{ij}}{\sum_k A_{ik}} = \frac{A_{ij}}{s_i}$$

where $s_i = \sum_k A_{ik}$ is the *strength* of node i.

Degree & Strength

Unweighted, undirected:

$$k_i = \sum_j A_{ij} \quad \text{(degree)}$$

Weighted:

$$s_i = \sum_j A_{ij}\, w_{ij} \quad \text{(strength)}$$

Directed:

$$k_i^{\text{in}} = \sum_j A_{ji}, \quad k_i^{\text{out}} = \sum_j A_{ij}$$

Structural Concepts

Paths and Connectivity

- A **path** $i_0 \to i_1 \to \cdots \to i_\ell$ exists if each edge is in E.
- G is **(strongly) connected** if every ordered pair is linked by a directed path.
- Strongly connected $\Leftrightarrow$ A is *irreducible*.

Spectral Properties

The **spectral radius** of A:

$$\lambda_{\max} = \max\{|\lambda| : \lambda \in \sigma(A)\}.$$

By Perron–Frobenius (non-negative, irreducible A), $\lambda_{\max}$ is a simple eigenvalue with a positive eigenvector.

$\lambda_{\max}$ will reappear as the epidemic threshold.

The network abstraction separates **structure** from **dynamics** — and reveals that the same structural quantities govern both.

A Third Application: Ranking Nodes by Importance

Problem. Given a directed network (e.g. the Web), assign a score $x_i > 0$ to each node reflecting its *importance*.

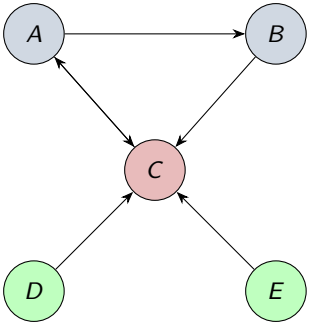
Naive answer: count how many nodes point to you.

$$x_i^{\text{naive}} = k_i^{\text{in}} = \sum_j A_{ji}.$$

But: a link from an unimportant page and a link from a high-traffic hub should *not* carry equal weight.

Better idea

A node is important if **important** nodes point to it.



C has the highest in-degree (4).
But A is backed by C, which is itself important.

The Recursive Definition and Eigenvector Centrality

Write the self-consistent condition explicitly:

$$x_i \propto \sum_j A_{ji}\, x_j,$$

i.e. the importance of i is the sum of the importances of all nodes that point to i.

In matrix form (proportionality constant λ):

$$A^\top \mathbf{x} = \lambda\, \mathbf{x}.$$

Eigenvector Centrality

$\mathbf{x}^*$ is the **leading eigenvector** of $A^\top$ (left eigenvector of A), normalised so $\|\mathbf{x}^*\|_1 = 1$. By Perron–Frobenius (irreducible A): unique and strictly positive.

Power Iteration

Compute $\mathbf{x}^*$ iteratively:

1. Start: $\mathbf{x}^{(0)} = \frac{1}{n}\mathbf{1}$.
2. Update:

$$\mathbf{x}^{(t+1)} = \frac{A^\top \mathbf{x}^{(t)}}{\|A^\top \mathbf{x}^{(t)}\|_1}.$$

3. Repeat until convergence.

This is *exactly* the same power iteration as for the stationary distribution — applied to $A^\top$ instead of P.

PageRank: Handling the Real Web

Pure eigenvector centrality breaks down on the real Web:

- **Dangling nodes**: pages with no out-links drain all importance.
- **Disconnected components**: Perron–Frobenius no longer guarantees a unique positive eigenvector.

Google's fix — the random surfer:

- Prob. d: follow a uniformly random out-link.
- Prob. $1-d$: teleport to a uniformly random page.

The **Google matrix** ($M_{ij} = A_{ij}/k_i^{\text{out}}$):

$$G = d\,M + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top.$$

PageRank = Stationary Distribution

G is irreducible and aperiodic by construction.

Therefore the **PageRank vector** π^{PR} is the *unique* stationary distribution of the random surfer:

$$\pi^{\text{PR}} G = \pi^{\text{PR}}.$$

Computed — again — by power iteration.

The same framework

PageRank is exactly the population-flow equilibrium from Section 2, applied to a cleverly constructed Markov chain on the Web graph. The network structure A determines everything.

A Second Application: Epidemics

Question. A pathogen spreads through a contact network. Under what conditions does it establish itself?

SIS Model (Susceptible–Infected–Susceptible)

Each node i is either **Susceptible** (S) or **Infected** (I).

- An infected neighbour j infects i at rate $\beta > 0$ per contact.
- An infected node recovers at rate $\gamma > 0$ (returning to S).

Let $\rho_i(t) = \Pr(\text{node } i \text{ is infected at time } t)$.

Mean-Field Equations on a Network

Under the **heterogeneous mean-field** (pair-independence) approximation:

$$\boxed{\frac{d\rho_i}{dt} = -\gamma\,\rho_i + \beta\,(1-\rho_i)\sum_{j=1}^{n} A_{ij}\,\rho_j, \quad i = 1, \ldots, n.}$$

Interpretation of each term:

- $-\gamma\rho_i$: recovery of node i.
- $(1-\rho_i)$: i must be susceptible to get infected.
- $\beta\sum_j A_{ij}\rho_j$: infection pressure from neighbours.

Linear Stability at $\rho \equiv 0$

Linearise: set $1-\rho_i \approx 1$.

$$\dot{\boldsymbol{\rho}} = (-\gamma I + \beta A)\,\boldsymbol{\rho}.$$

The disease-free state is **unstable** iff the matrix $\beta A - \gamma I$ has a positive eigenvalue, i.e.

$$\beta\,\lambda_{\max}(A) - \gamma > 0.$$

The Epidemic Threshold

Epidemic Threshold on Networks

Define the **effective spreading rate** $\tau = \beta/\gamma$. Then:

$$\tau_c = \frac{1}{\lambda_{\max}(A)}.$$

- $\tau < \tau_c$: the infection decays exponentially — **disease-free phase**.
- $\tau > \tau_c$: a non-trivial endemic state exists — **epidemic phase**.

Structural insight

$\lambda_{\max}(A)$ grows with the number and strength of connections. **Hubs** (high-degree nodes) raise $\lambda_{\max}$ and *lower* the threshold — they are disproportionately dangerous as super-spreaders.

Comparing Homogeneous and Heterogeneous Networks

Homogeneous network (all degrees $\approx \langle k \rangle$):

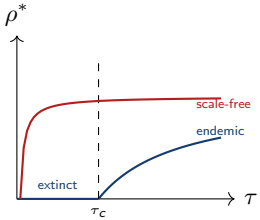
$$\lambda_{\max}(A) \approx \langle k \rangle \implies \tau_c \approx \frac{1}{\langle k \rangle}.$$

Matches the classical SIS result on a regular graph.

Scale-free network ($k \sim k^{-\alpha}$, $\alpha \in (2,3)$):

$$\lambda_{\max}(A) \sim \sqrt{k_{\max}} \gg \langle k \rangle \implies \tau_c \to 0 \text{ as } n \to \infty.$$

Epidemics spread for *any* non-zero τ!



Phase diagram: endemic fraction ρ^* vs. spreading rate τ.

Summary

What we covered

1. **Methodology**: example → trial & error → theory → network.
2. **Markov chains**: transition matrices, stationary distributions, Perron–Frobenius.
3. **Networks**: vertices, edges, weights, degree, adjacency matrix, spectral radius.
4. **Epidemics**: SIS mean-field on networks, epidemic threshold $\tau_c = 1/\lambda_{\max}$.

The unifying thread

The **spectrum of the network matrix** governs long-run behaviour across very different dynamical systems.

Further reading:

Newman, *Networks* (2018).
Pastor-Satorras & Vespignani, *Phys. Rev. Lett.* **86** (2001).

Thank you.